

畅翔宇

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个人主页: https://changxiangyu97.github.io/Xiangyu_Chang/



教育经历

安徽建筑大学, 道路桥梁与渡河工程, 本科	2014.09 - 2018.06
东南大学, 工程力学, 硕士	2018.09 - 2021.06
东南大学, 研究助理	2021.06 - 2022.07
南洋理工大学, 土木工程, 博士	2022.08 - 2026.08

学术论文

关键词: Digital Twin, Underground Structure, Smart City, Computer Vision, Surrogate Strategy

主要成果:

1. **Xiangyu Chang**, Hongyun Fan, et al. 2026. Domain-Adaptive Digital Twins for Multi-Task Vibration-Based Damage Assessment of Shield Tunnels. *Automation in Construction*, 188, 106988. (IF = 12.6)
2. **Xiangyu Chang**, Hongyun Fan, et al. 2026. A Transformer-Based Surrogate Modeling Strategy for Tunnel Digital Twin in Full-field Displacement Prediction under Adjacent Tunnel Construction. *Advanced Engineering Informatics*, 69, 104045. (IF = 11.5)
3. **Xiangyu Chang**, Youqi Zhang, et al. 2026. Structural Evaluation of Cracked Shield Tunnels Using Computer-vision-based Model Updating Techniques. *Advanced Engineering Informatics*, 69, 103818. (IF = 11.5)
4. **Xiangyu Chang**, Rui. Zhang, et al. 2024. Digital Twins in Transportation Infrastructure: An Investigation of the Key Enabling Technologies, Applications, and Challenges. *IEEE Transactions on Intelligent Transportation Systems*, 25(7), 6449-6471. (IF = 9.1)
5. **Xiangyu Chang**, Hao Wang, and Yiming Zhang. 2023. Back Analysis of Rock Mass Parameters in Tunnel Engineering Using Machine Learning Techniques. *Computers and Geotechnics*, 163. (IF = 7.1)
6. **Xiangyu Chang**, Hao Wang, Yiming Zhang, et al. 2021. Bayesian Prediction of Tunnel Convergence Combining Empirical Model and Relevance Vector Machine. *Measurement*, 188. (IF = 6.1)
7. **Xiangyu Chang**, Rui Zhang, et al. 2025. Modeling and Quantitative Assessment of Critical Defects in Shield Tunnels with Cascading Effects. *Geotechnical and Geological Engineering*, 43, 194. (IF=3.2)
8. **Xiangyu Chang**, Hongyun Fan, et al. 2026. Experimental Study on the Dynamic Response Characteristics of Cross Twin Tunnels Under Subway Train Load. *Geotechnical and Geological Engineering*, 44, 381. (IF=3.2)
9. **Xiangyu Chang**, Youqi Zhang, Yuguang Fu. 2024. A Computer-vision-based Model Updating Strategy for Shield Tunnels with Cracks. *World Tunnel Congress 2024*. (Conference paper)
10. Yishun Xie, **Xiangyu Chang**, Jianxiao Mao, et al. 2023. Prediction and Early Warning of Extreme Winds for High-Speed Railway Bridge Construction Using Machine-Learning Methods. *Sustainability*, 15(24), 16921. (IF=4.1)
11. Yintao Chen, Xin Shao, **Xiangyu Chang**, et al. 2025. Prediction of Surrounding Rock Deformation in a

- Highway Tunnel Using an LSTM-RF Hybrid Model. *Journal of Engineering*, 6652758. (Corresponding author) (IF=2.6)
12. Zhaozhong Li, **Xiangyu Chang**, Hao Wang, et al. 2023. Back-analysis Method of Rock Mass Properties in Tunnel Engineering Using Multiple Monitoring Data Based on LS-SVR Algorithm. *Journal of Southeast University (English Edition)*, 1:1-7.
13. 王浩, **畅翔宇**, 张一鸣, 等. 高速铁路隧道围岩力学参数反演研究. *铁道工程学报*, 2020, 37(9): 47-53.

相关成果:

1. Hongyun Fan, Liping Li, Yuguang Fu, **Xiangyu Chang**. 2025. 3D DDA-MPM Coupling Method for Fluid-solid Interaction: Theory, Validation and Application in Tunnels. *Computers and Geotechnics*, 70166.
2. Hongyun Fan, Liping Li, Yuguang Fu, Hongliang Liu, **Xiangyu Chang**. 2025 Block Collapse Prediction and Reinforcement Optimization in Tunnels Based on Discontinuous Deformation Analysis and Machine Learning Models. *International Journal for Numerical and Analytical Methods in Geomechanics*.
3. Shiqi Wang, **Xiangyu Chang**, et al. 2024. Off-design Performance Optimization for Steam-water Dual Heat Source ORC Systems. *Process Safety and Environmental Protection*, 191.
4. Yiming Zhang, Hao Wang, Yu Bai, Jianxiao Mao, **Xiangyu Chang**, et al. 2021. Switching Bayesian Dynamic Linear Model for Condition Assessment of Bridge Expansion Joints Using Structural Health Monitoring Data. *Mechanical Systems and Signal Processing*, 160107879.
5. Shiqi Wang, Zhongyuan Yuan, Kim Tiow Ooi, **Xiangyu Chang**, et al. 2025. Optimization of Performance under Off-design Conditions for Dual-pressure Organic Rankine Cycle with Hot Source Splitting. *Energy Nexus*, 17.
6. 李照众, 王浩, **畅翔宇**, 等. 基于支持向量回归优化的高铁隧道围岩收敛变形预测研究. *东南大学学报(自然科学版)*.

发明专利和软件著作权

1. 王浩, **畅翔宇**, 梁瑞军, 等. 一种基于神经网络技术的超前小导管设计方法. 发明专利, 2019
2. 王浩, **畅翔宇**, 祝青鑫, 等. 一种隧道初期支护钢拱架安装装置及使用方法. 发明专利, 2020
3. 王浩, **畅翔宇**, 刘奕辰, 等. 一种土压力盒安装装置及安装方法. 发明专利, 2021
4. 张一鸣, 王浩, **畅翔宇**. 耦合数值天气预报与实测数据的风速多点同步预测方法. 发明专利, 2022
5. 王浩, 何祥平, **畅翔宇**. 高铁隧道施工安全监测及分析软件[简称: SMAHT]. 软件著作权, 2019

参与科研项目

1. 国家万人计划青年拔尖人才项目. W03070080. (参与)
2. 江苏省重点研发计划(产业前瞻与共性关键技术)项目. BE2018120. (参与)
3. 跨断层高铁隧道施工智能监控及安全预警研究. 横向课题. (主要负责人)
4. 大角度倾斜箱涵顶进施工智能控制及风险预警技术研究. 横向课题. (参与)

5. A smart digital twin framework using advanced modelling and data analytics for monitoring and management of underground transportation infrastructure. Ministry of Education Tier 1 Seed Fund. (参与)
6. Novel context-aware multivariate time series modelling for underground transportation infrastructure monitoring and management. AI Singapore. (参与)

个人奖励

大学生数学建模竞赛（二等奖）	2015
国家励志奖学金	2015
三好学生	2015
Mathematical Contest In Modeling (MCM/ICM) (三等奖)	2016
优秀毕业生	2018
一等、二等奖学金 (8次)	2014-2018
先进个人	2019
三好学生	2020
东大设计院奖学金	2020
南洋理工大学奖学金	2022

社团和组织经历

党支部联合中心	2018.9-2020.6
部门负责人 创新部	
在东南大学党支部联合中心任职，主要协助学校研工部处理党建工作以及研究生数学建模竞赛秘书处工作。	
基层党建工作	2018.11-2020.12
党支部书记	
担任2018级硕士第五党支部党支书，负责支部党日活动、党员发展等事务，组织党日活动。任职期间共有4名同志成为发展对象、8名积极分子。	
南洋理工大学助教	2023.11-2025.12
本科生助教老师	
担任CV0003 Introduction to Data Science and Artificial Intelligence 和 CV2712 Civil Engineering Laboratory两门课程的老师，负责课程讲解、答疑，实验助理以及考试安排等。	